

```
pip install transformers torch sentencepiece
```

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Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.7.1)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.0)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0)
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0)
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Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0)
```

```
from transformers import pipeline
```

```
print("Transformers installed successfully!")
```

```
Transformers installed successfully!
```

```
# Load summarization pipeline with pretrained model
summarizer = pipeline("text-generation", model="facebook/bart-large-cnn")

print("Model loaded successfully!")
```

```
model.safetensors: 100% 1.63G/1.63G [00:12<00:00, 181MB/s]
Please make sure the generation config includes `forced_bos_token_id=0`.
Loading weights: 100% 316/316 [00:00<00:00, 672.82it/s, Materializing param=model.decoder.layers.11.self_attn_layer_norm.weight]
BartForCausalLM LOAD REPORT from: facebook/bart-large-cnn
Key | Status |
-----+-----+
model.encoder.layers.{0...11}.self_attn.v_proj.weight | UNEXPECTED |
model.encoder.layers.{0...11}.final_layer_norm.bias | UNEXPECTED |
```

```
# Different domain texts
texts = [
    # Technology
    """Artificial Intelligence is a rapidly growing field that focuses on creating intelligent machines.
    It has applications in robotics, natural language processing, and computer vision."""

    # Healthcare
    """Healthcare systems are improving with the use of AI technologies. Doctors use machine learning
    to detect diseases early and provide better treatment."""

    # Education
    """Online education platforms are making learning accessible to students worldwide.
    They provide flexible and personalized learning experiences."""
]

# Generate summaries
for i, text in enumerate(texts):
    summary = summarizer(text, max_length=50, min_length=20, do_sample=False)
    print(f"\nText {i+1} Summary:")
    print(summary[0]['generated_text'])
```

```
merges.txt: 456k/? [00:00<00:00, 12.0MB/s]
Text 1 Summary:
Artificial Intelligence is a rapidly growing field that focuses on creating intelligent machines.
It has applications in robotics, natural language processing, and computer vision. still had - still - then - had - only bot
Model loaded successfully!

Text 2 Summary:
Healthcare systems are improving with the use of AI technologies. Doctors use machine learning
to detect diseases early and provide better treatment. had had still had had had " had had only had had both had had were ha

Text 3 Summary:
Online education platforms are making learning accessible to students worldwide.
They provide flexible and personalized learning experiences. only only only had had had only had only only one only only sti
```